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Ethics

One ethical consideration that we had was one of accuracy. We were providing information about stocks and the way that we did it we would be ethically and legally liable for providing false information. Lying or not telling the truth is an inherently ethical thing and the responsibility would be on us, if people trusting our information made false trades then they could sue us for the loss that they had. Section 5 of the IEEE code of ethics is this “to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, to be honest and realistic in stating claims or estimates based on available data, and to credit properly the contributions of others;”

In response to this we tried to have the most accurate information as we could get. Repeatedly reviewing methods and calculations to try to be as accurate as possible. We also checked for each value how often that value would change and changed them on those time frames. In addition we also tried to be as clear as possible about how each value was calculated. We had a description for each of the values in the stock table.

Problem

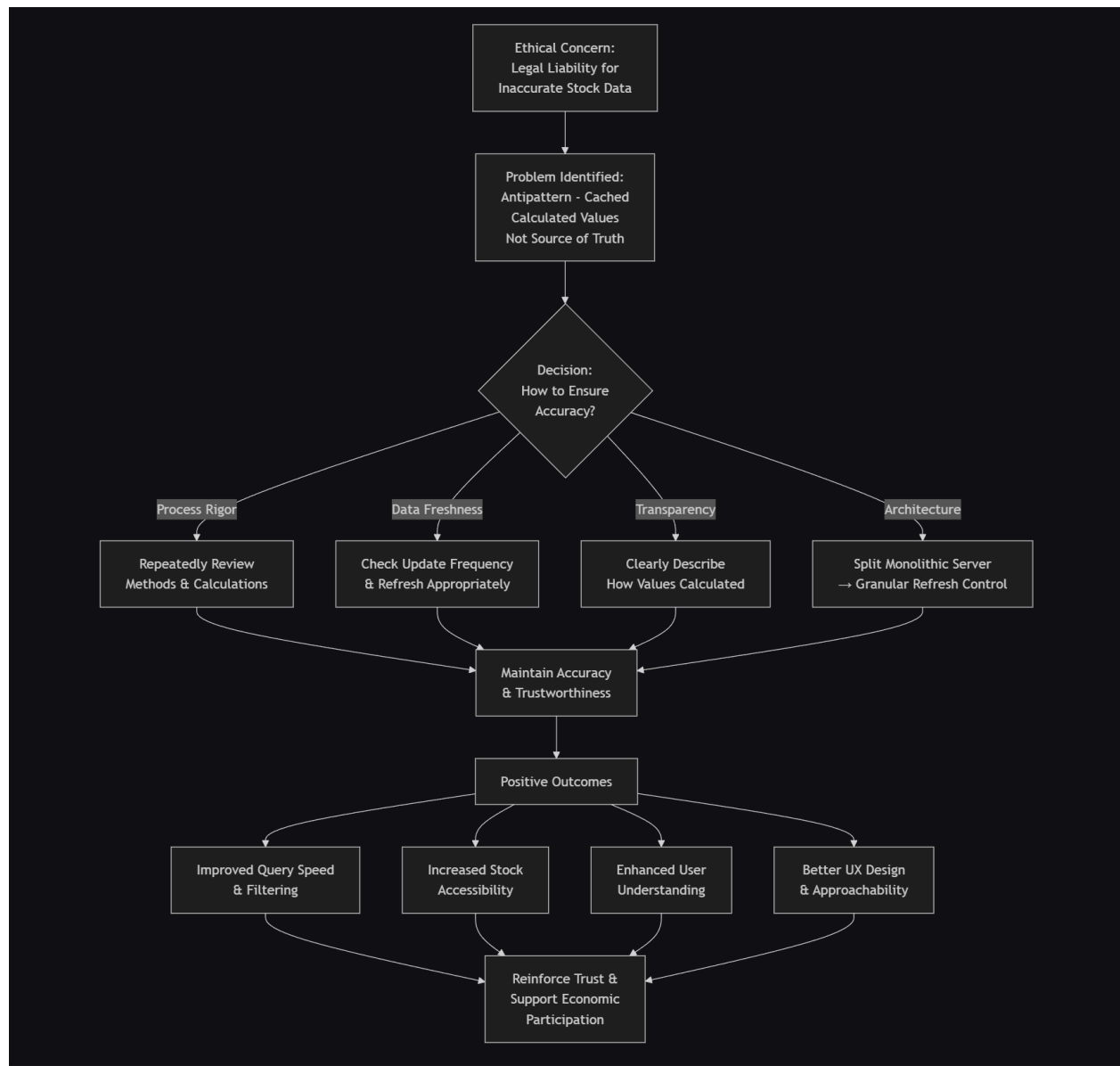
For a different problem our solution was by using the antipattern of saving certain calculated values in our database. This was to increase the speed of the queries which

were incredibly slow when they included calculations of data. This is problematic, however, due to the data that we give to the consumer not being the source of truth

It was then that we needed to make sure the data we were storing was up to date. Originally to solve this we had one mega server that handled fetching new stock data, storing it in the database, as well as calculating that data and storing the calculations in the database. We decided to split this up to give us more granular control over how often we refreshed this data.

Area of consideration

The primary consideration of this app would have to be economic given that it is a stock investment advisor app. The primary Goal or mission statement of the app would be “To increase understanding in stock investment such that we give the power back to the people”. This has the economic effect of strengthening the economy and allowing more popular things to gain funding. This is primarily economic but secondarily social. A couple of the ways that we have tried to do this is to increase the quality and focus of the app to be more educational as opposed to functional. To do this we did things mentioned previous such as have informational tabs for the values that we had, and a help page to manage searching the stocks. Even though the model wasn't the most accurate we also had a neural network that predicted stock information. But the constraint on time prevented us from getting it to where we would have liked.



Relevant codes of ethics or engineering standards referenced

IEEE code of ethics